Printed holders: what goes where

Every tower gets the same four holders. Each holder carries one device that is taped to the tower now; the chopper, ESCON and motor unit stay where they are.

Printed part (file)	Device it carries	Device held by	Holder fixed to tower by
DIN-rail mount (iaa100_din_mount.stl)	Futek IAA100 force amplifier (small black box, load-cell cable)	nothing: the IAA100 clicks onto the rail	2 x M6x16 + 2 T-nuts, one slot
24 V plate (rs15_plate.stl)	Mean Well RS-15-24 (small 24 V supply)	2 x M3x6 from the back	2 x M6x16 + 2 T-nuts, one slot
48 V brackets A + B (rsp2000_brackets.stl; B alone: rsp2000_bracket_B.stl)	Mean Well RSP-2000-48 (large 48 V supply, 1.95 kg)	A: 2 x M4x12, B: 1 x M4x12, from the back	2 x M6x16 + 2 T-nuts each, both slots of one face
Ground bar plate (ground_bar_mount.stl)	ONPIRA V 1x12 distributor (12 screw terminals, all connected): the tower's ground wires	2 x M3x12 + 2 M3 nuts	2 x M6x16 + 2 T-nuts, one slot

Per tower / for all 4 towers

Printed: 1 DIN-rail mount, 1 24 V plate, 1 pair of 48 V brackets, 1 ground bar plate / 4, 4, 4 pairs, 4.
Plus once, on the tower with the carrier board: 1 carrier board plate, 2 x M6x16, 2 T-nuts, 4 x M3x16 + 4 M3 nuts.
Screws: 10 x M6x16 button head, 10 slot-10 T-nuts, 2 x M3x6, 3 x M4x12, 2 x M3x12 + 2 M3 nuts / 40, 40, 8, 12, 8 + 8.

Before you start

1. Power: the tower's power strip unplugged from the wall. Mains wiring (230 V side of both supplies) stays with the electrician.
2. T-nuts: slide-in T-nuts only go in from the end of the profile. If the profile ends are blocked (motor unit on top, floor mount at the bottom), use drop-in (swivel / hammer-head) slot-10 T-nuts: they go in from the front and turn 90 degrees when tightened.
3. Fit: done with the fit-check print (2026-10-22): the key (0.2 mm clearance) slides into the slot without play. Keep it for all parts.
4. Always fit the device to its holder first, then the holder to the tower (the carrier board plate is the other way round: page 7).

Words used

1. Key: the 9.6 mm ridge on the back of each holder. It sits in the profile slot and stops the holder from turning.
2. Slot: the groove along each face of the aluminium profile. Each 90 mm face has two slots, 45 mm apart.
3. Face: one of the four flat sides of the tower profile.

What to print now

Fit-check print (2026-10-22): IAA100 mount and RS-15 plate fit the devices, 48 V bracket A fits the supply's cable end. Changes since then: the RS-15 plate is longer (M6 holes moved out: one hit the end stop, the other sat under the mains cables); bracket B (fan end) was never in the fit check and is now 20 mm longer past the supply, so its T-nuts have room.

One file: out/tower_set.stl, the 5 parts of one tower on one bed (235 x 166 mm), laid out like the fit check. Print it 4 times, once per tower (the fit-check A and IAA100 mount become spares).

Part (in tower_set.stl)	Per set	Size (mm)	Note
rs15_plate	1	128 x 51 x 14	new version; the fit-check RS-15 plate is not used
rsp2000_bracket_A	1	62 x 107 x 18	48 V, cable end; unchanged
rsp2000_bracket_B	1	82 x 107 x 18	48 V, fan end; ASA, or PETG with 40 %+ infill
iaa100_din_mount	1	100 x 40 x 15	unchanged
ground_bar_mount	1	117 x 32 x 14	new: ONPIRA V 1x12 ground bar; page 8
carrier_plate.stl (own file)	1 in total	204 x 86 x 18	only for the tower with the carrier board; page 7

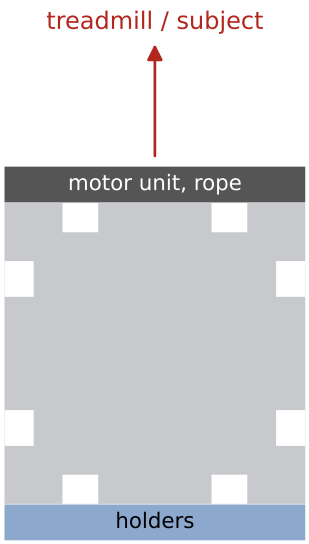
Print settings (same as the fit check, which printed fine)

1. Flat, key side down, exactly as exported: no part is taller than 18 mm. Do not stand parts on end (that toppled 4 times, 2026-10-19).
2. Supports touching the build plate, under: the plate wings (2.5 mm above the bed, every part), the 48 V brackets' pad ledges (10.5 mm beyond the plate at both sides, 7.5 mm above the bed) and the DIN rail flanges (14 mm above the bed). Small round pockets on the bottom (screw-head recesses, 6-8 mm) bridge without support.
3. PETG: support Z gap 0.2-0.25 mm (PETG welds to its supports; the 1 mm DIN rail flanges break if supports are pried off hard), 4 walls, 40 % or more infill. On a smooth PEI bed use glue stick as a release layer.
4. Do not scale and do not thicken the key, the 5 mm plate under the M6 heads or the hole positions: they set the fit.
5. PETG or ASA, not PLA for the 48 V brackets (PLA softens around 55-60 C, on a warm supply). PETG holds the 1.95 kg easily; its limit is heat (softens around 70 C): touch the brackets after the first long session, they should feel only warm.

Where on the tower (recommended)

Heights are not fixed: put each holder where its cables reach without strain. If a device is already taped at a spot that works, put its holder right there. The layout below is the recommended starting point.

Seen from above



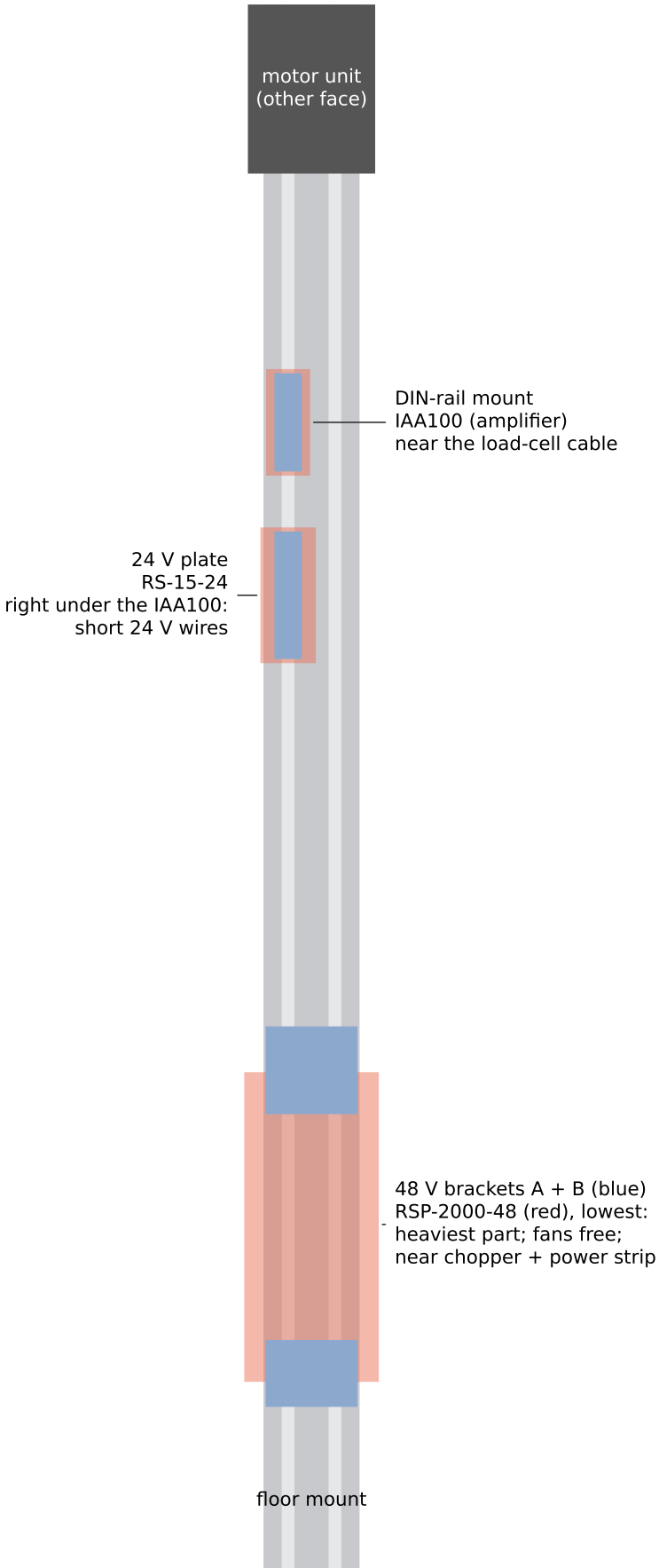
back face: electronics

Put the holders on the face pointing away from the treadmill: the rope side stays clear and nobody on the treadmill can touch the 48 V parts. A side face is fine if the back is blocked.

Rules behind the layout

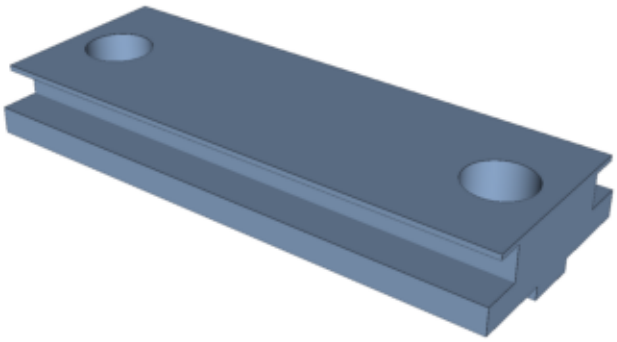
1. IAA100: within reach of the load-cell cable (the load cell hangs in the rope near the eyebolt), and of the 5-wire signal cable.
2. RS-15-24 next to the IAA100: its +V / -V go to IAA100 pins 1 / 2. Its 230 V terminals face away from hands; the electrician covers them.
3. IAA100 and RS-15 use the same slot, one above the other (side by side they would overlap by 0.5 mm). The RS-15 plate is 128 mm long.
4. RSP-2000 low on the tower: 1.95 kg, close to the power strip and to the chopper (its +V / -V go to chopper pins 4 / 5). Hang it along the profile, both brackets in both slots, fans at the ends not blocked, 10 mm air gap behind.
5. Leave a hand's width free around the chopper and ESCON: they get warm.

Back face from behind
(heights not to scale)

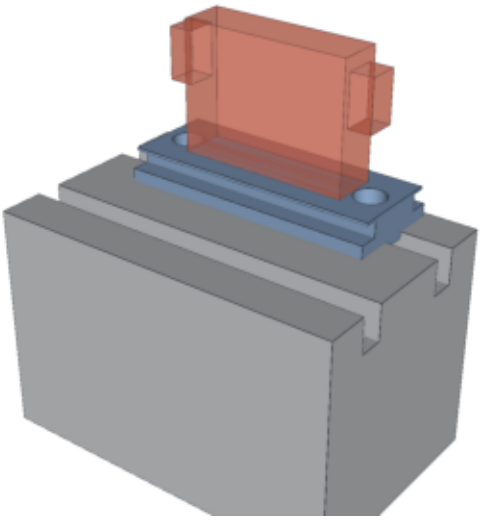


1. IAA100 on the DIN-rail mount

The printed part



On the profile (grey), device (red)



Holds the Futek IAA100 force amplifier (now taped) on the tower. The IAA100 has a spring DIN clip on its back; this part is a short 35 mm DIN rail that bolts into one slot, so the amplifier clicks on and off without screws.

Steps

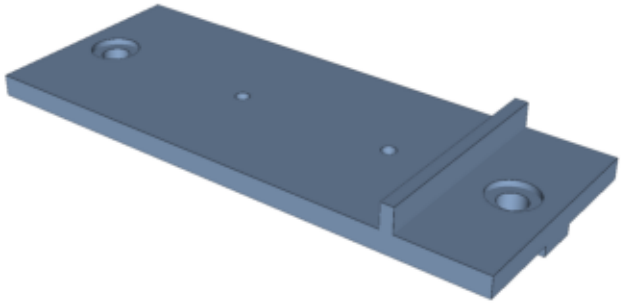
1. Choose a slot on the tower face near the load cell cable; slide 2 slot-10 T-nuts into it, about 80 mm apart.
2. Put the mount on with the key in the slot and the rail facing out; bolt it with 2 M6 bolts through the holes at the ends (reach them through the openings in the rail).
3. Hook the top of the IAA100's DIN clip over the rail, press the bottom until it clicks.
4. Orientation: sensor plug (4-pin) toward the load cell, power/output plug (5-pin) toward the breadboard cable.
5. To remove: pull the clip's spring tab down (small screwdriver) and tilt the amplifier off.

Check

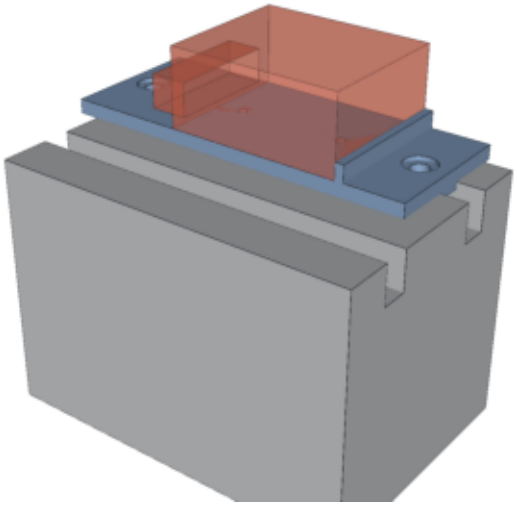
Push the amplifier: it does not move. Load-cell cable and signal cable reach without pulling.

2. RS-15-24 on the 24 V plate

The printed part



On the profile (grey), device (red)



Holds the Mean Well RS-15-24, the 24 V supply of the force amplifier (now taped). Fixed by the supply's own 2 M3 holes; an end stop keeps it square.

Steps

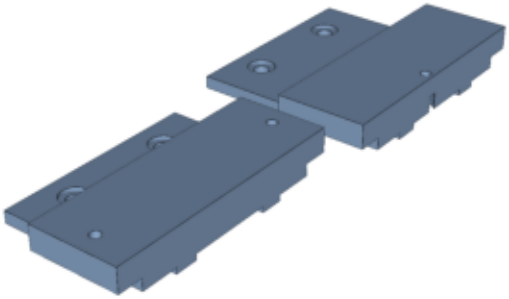
1. Mains first: the supply stays unplugged; its 230 V terminals are wired and covered only by the electrician.
2. Put the supply on the plate, the end stop against the end opposite the screw terminals.
3. From the back of the plate, screw 2 M3 x 6 screws into the supply's bottom holes (max 3 mm into the case: no longer screws).
4. Slide 2 slot-10 T-nuts into a slot, 128 mm free; bolt the plate on with 2 M6 bolts through the holes at both ends (the one at the terminal end is 25 mm past the supply, away from the mains cables).
5. Route the 24 V output (+V, -V) to the IAA100 power plug (pin 1 +Vin, pin 2 GND).

Check

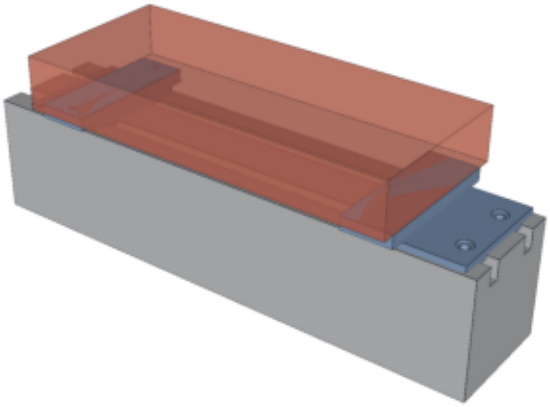
Supply sits flat against the end stop; screws not longer than M3x6 (max 3 mm into the case). Both M6 heads clear: the mains cables do not touch the one at the terminal end. 230 V terminals covered before the strip is plugged in.

3. RSP-2000-48 on the two brackets

2 printed parts: bracket A (2 small holes) and B (1 small hole)



On the profile (grey), device (red)



Hold the Mean Well RSP-2000-48, the motor supply (1.95 kg), on one profile face with a 10 mm air gap. Bracket A takes the 2 M4 holes at the end without fans, bracket B the single M4 hole at the fan end.

Steps

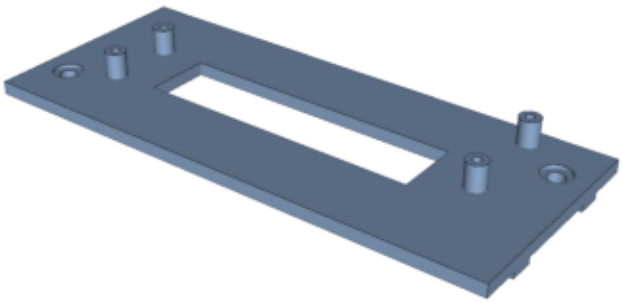
1. Supply switched off and unplugged; leave its mains side to the electrician.
2. Lay the supply with its wide face (the one with 3 M4 holes, case 952D) up. Put bracket A on the end with 2 holes and bracket B on the fan end, pads on the supply, the bolt flanges pointing away from the supply.
3. Screw M4 x 12 screws from the back of each bracket into the supply (2 in A, 1 in B); max 4 mm into the case. B's screw goes into the hole near the middle of the fan end (34.5 mm from the end); the 2 small holes beside it are not for mounting.
4. Slide 4 slot-10 T-nuts into the tower face: one in each of its 2 slots for each bracket. The two brackets' bolts end up 339 mm apart.
5. Lift supply + brackets onto the profile, keys in both slots, and bolt each bracket with its 2 M6 bolts. Two people: one holds. Tighten firmly with the short end of the hex key only (about 2 N m): more crushes the plastic under the head.
6. Check: fans and vents free, 10 mm gap behind the supply, output cables (+V, -V) reach the chopper terminals 4 / 5.

Check

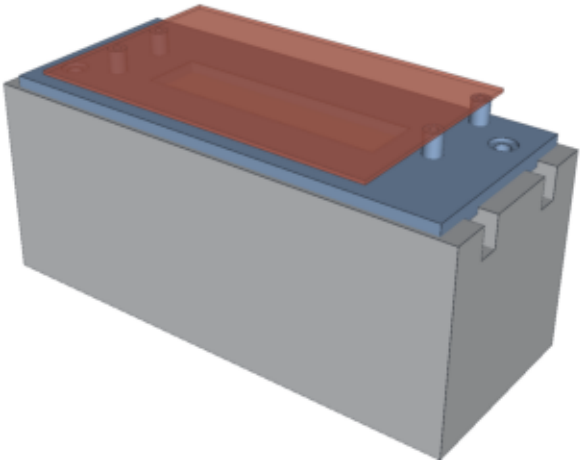
Two people. Push the supply: nothing moves, keys sit in both slots. Fans and vents free, 10 mm gap behind. Output cables reach chopper pins 4 / 5 without strain.

4. Carrier board on its plate (one tower)

The printed part (4 standoffs)



On the profile (grey), device (red)



Holds the carrier board (Teensy 4.1, DAC and the 4 module plugs) on one face of the tower that carries it, long side along the profile, on 10 mm standoffs so the soldered pins underneath touch nothing.

Steps

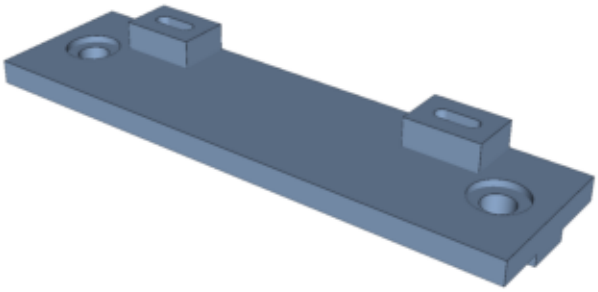
1. Push an M3 nut into each of the 4 hexagon pockets in the back of the plate (a drop of glue or tape keeps them in).
2. Slide 2 slot-10 T-nuts into the face, one in each slot, 180 mm apart (the plate's 2 M6 holes sit diagonally).
3. Bolt the plate on, keys in both slots, with 2 M6 bolts; tighten with the short end of the hex key.
4. Put the board on the 4 standoffs, component side out: they only fit one way round (2 holes at one end, 2 at the other). Screw 4 M3 x 16 through the board into the nuts, gently: plastic underneath.
5. Teensy USB end up or down as the cables run best: turn the plate 180 degrees before bolting it on. Module plugs face one side: route the 4 module cables and the USB cable without strain.

Check

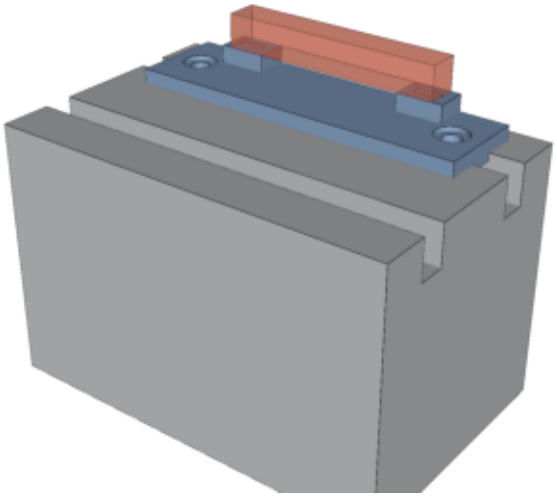
Board does not rock; no solder joint underneath touches the plate (10 mm gap). USB and the 4 module plugs reachable, cables without strain. Both M6 heads can be reached past the board ends.

5. Ground bar (ONPIRA V 1x12)

The printed part (slots for any hole spacing 62-74 mm)



On the profile (grey), device (red)



Holds the ONPIRA V 1x12 distributor: 12 screw terminals, all connected, so every ground wire of the tower (ESCON, IAA100, supplies' -V, module cable) lands on one bar instead of a twisted bundle.

Steps

1. Slide an M3 nut into each of the 2 channels in the back of the plate (flat side up): it slides along, it cannot turn.
2. Put the board on the 2 pads (terminal screws up, beside the key, not over it). Line each nut up under the board's hole; push an M3 x 12 through the board into it and tighten gently. The included 5 mm sleeves and wood screws are not needed.
3. Slide 2 slot-10 T-nuts into a slot, 97 mm apart; bolt the plate on with 2 M6 bolts at its ends.
4. Wire the grounds; mark the bar GND. No 230 V protective earth on this bar (that stays with the electrician).

Check

Bar does not move; both M3 screws found their nuts (if a hole is outside the slot, measure the hole spacing and tell Claude). Ground wires reach without strain.